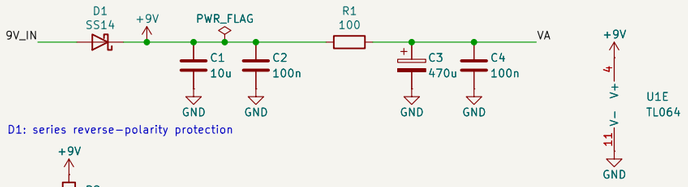
## POWER & BIAS



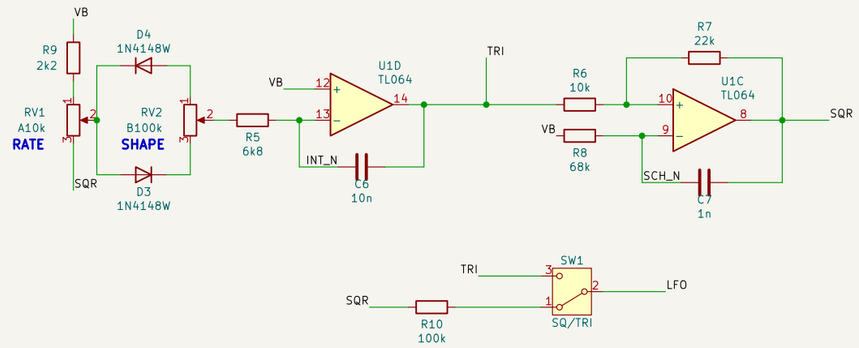




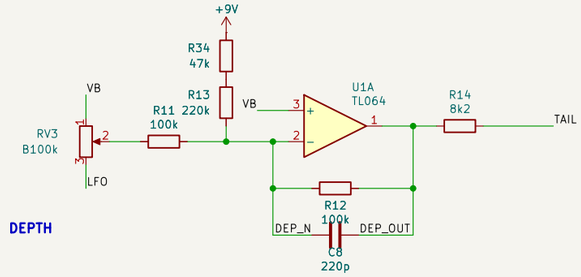
VA = filtered supply for the VCA/output

VB = 4.5V bias (U1B buffer)

## LFO (U1C Schmitt + U1D integrator)

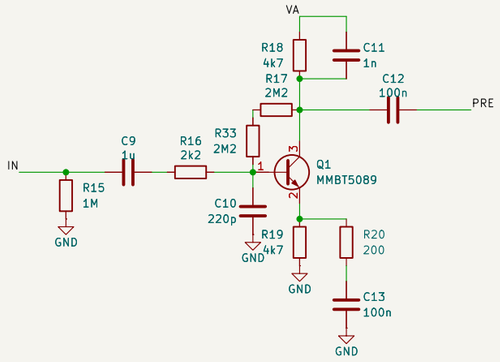



## DEPTH (U1A)

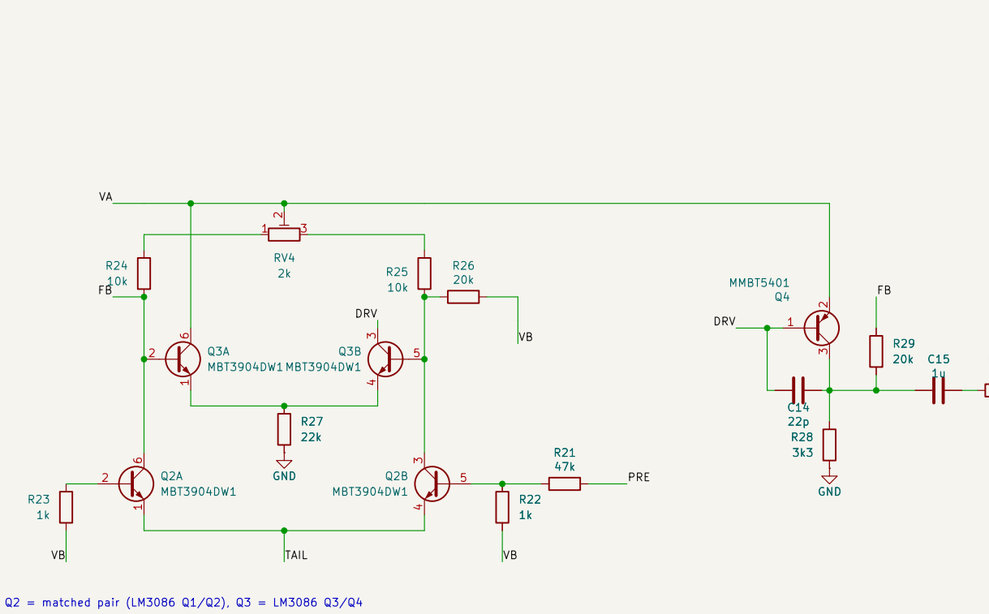



U1A converts the LFO into the VCA tail current

## INPUT PREAMP (Q1)



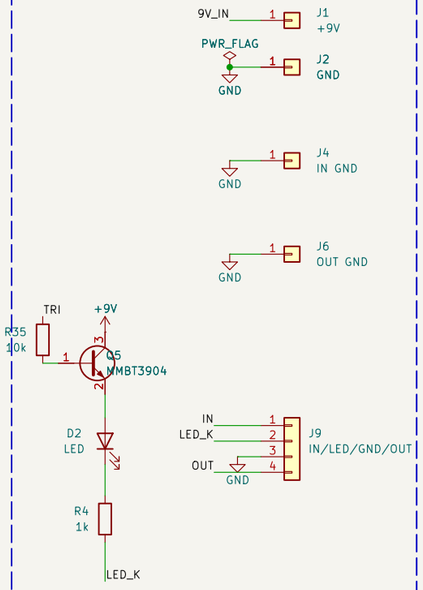

## VCA & OUTPUT (discrete LM3086: Q2, Q3 + Q4)



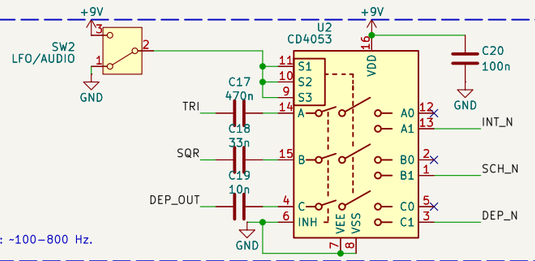

Q2 = matched pair (LM3086 Q1/Q2), Q3 = LM3086 Q3/Q4

RV4 balance trim (v2.1): 3296W 2k multi-turn, wiper to VA; null with RV4 near centre. R26 mirrors R29

## OFF-BOARD WIRING




## RANGE (U2 CD4053)




LFO: SW2 to +9V adds C17–C19 (~2–17 Hz). AUDIO: ~100–800 Hz.

## NOTES

1. SMD passives 0805 unless noted (C3 8x10.5mm electrolytic). LCSC field = JLCPCB part number.
2. Hand-soldered on the BACK side: RV1–RV3 (Omeg 16mm), SW1 (SQ/TRI toggle), D2 (3mm LED). J1–J8 are wire pads.
3. Changes from Trem-22-01: Q1/Q4(was Q6) SOT-23 pinout corrected; D5+R38 shunt replaced by series SS14 (D1); LM3086 -> 2x MBT3904DW1:

   C1/C5 10uF electrolytics -> 0805 MLCC; 470n and 100k parts swapped to in-stock basic parts.

Super-Freq.com
Converted from Eagle Trem-22-01 for JLCPCB SMD assembly

| | | |
|---|---|---|
| Sheet: / | | |
| File: Trem_23.kicad_sch | | |
| **Title: Trem 23 – SMD** | | |
| Size: A3 | Date: 2026-09-24 | **Rev: 2.1** |
| KiCad E.D.A. 10.0.4 | | Id: 1/1 |